

CIFAR-10 Object Recognition Using Convolutional Neural Networks

Track 2 Deep Learning Approach

Neural Network Models for Object Recognition Using Multi-Track ML Approaches



Research Objective

Project Aim

01 Architecture

Train and evaluate object recognition on CIFAR-10

02 Methodology

Separate **training**, **validation**, and **test** datasets

03 Optimization

Impact of hyperparameters and **data augmentation**

04 Metrics

Analyze performance using precision and recall

05 DEMONSTRATION & COMPARATIVE DISCUSSION



Dataset Overview

CIFAR-10 Benchmark

Benchmark image classification dataset

TOTAL SAMPLES

60,000

DIMENSIONS

32×32 px

CHANNELS

RGB (3)

CATEGORIES

10 Classes

DISTRIBUTION

6,000 per class

airplane

automobile

bird

cat

deer

dog

frog

horse

ship

truck

KRIZHEVSKY (2009); TENSORFLOW (2024)



Dataset Split

Data Partitioning

CIFAR-10 Standard Distribution

ORIGINAL TRAIN

50,000

ORIGINAL TEST

10,000

WORKING PROJECT PIPELINE

TRAINING

40,000

VALIDATION

10,000

TEST

10,000

The validation set was partitioned from the original training data as required by the assignment.



Methodology Deep-Dive

Why Validation Matters

TRAIN

Model Learning

VAL

Hyperparameter Tuning

TEST

Final Verification

- Training accuracy alone can be misleading
- Validation checks generalisation on unseen data
- Test set reserved for final evaluation only

Goodfellow, Bengio and Courville, 2016

Technical Note

Normalization: 0-255 → 0.0-1.0. Essential for stable optimization gradients.



Track 2: Deep Learning

CHOSEN APPROACH

Convolutional Neural Network

CNNs learn spatial features directly from raw image pixels, providing superior translation invariance.

More suitable for classification than classical methods relying on hand-crafted features.

LeCun et al., 1998

TESTED ARCHITECTURES

MODEL A

Baseline CNN architecture

MODEL B

Baseline + Data Augmentation



Baseline CNN Architecture

ARCHITECTURAL FLOW

Input: $32 \times 32 \times 3$ (RGB)

Conv2D: 32 Filters (3×3) + ReLU

MaxPooling2D (2×2)

Conv2D: 64 Filters (3×3) + ReLU

MaxPooling2D (2×2)

Flatten Layer

Dense: 64 Units + ReLU

Dense: 10 Units + Softmax

TRAINING HYPERPARAMETERS

Epochs

10

Batch Size

64

Optimizer

ADAM

Learning Rate

0.001

Activation Functions

- Hidden: Rectified Linear (ReLU)
- Output: Softmax (Multiclass)

Goodfellow, Bengio and Courville
(2016)



Baseline Training Results

TRAINING ACC

0.7681

VALIDATION ACC

0.6918

TEST ACCURACY

0.6945

ANALYSIS & INTERPRETATION

- Performance significantly exceeds random guessing (0.10 for CIFAR-10).
- Stable generalization across validation and test sets.
- Model A serves as a robust benchmark for Track 2 development.

OVERFITTING ALERT

The ~7.6% delta between Train and Val indicates moderate overfitting. Regularization or data augmentation is recommended for subsequent iterations.



Data Augmentation Experiment

Improving Generalisation

CONTROL: MODEL A

Baseline architecture serving as the performance benchmark.

- No data transformation
- Raw CIFAR-10 input
- Fixed spatial features

VARIABLE: MODEL B

Same CNN backbone with input-stage augmentation:

- Random horizontal flip
- Small random rotation
- Increased training variation

CONSTANT SETUP: 10 Epochs • 64 Batch Size • ADAM Optimizer • 0.001 Learning Rate

Shorten and Khoshgoftaar, 2019



Model Comparison

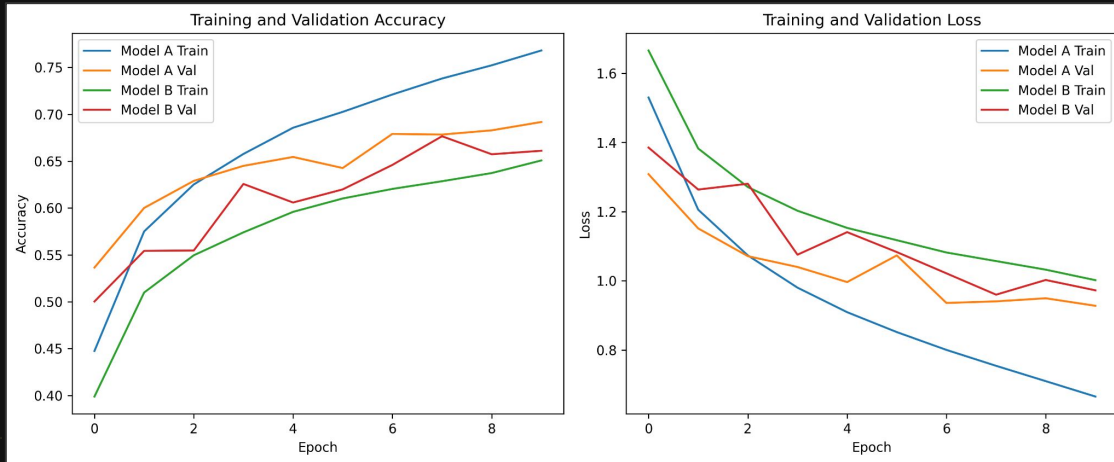
Model	Train Acc	Val Acc	Test Acc
Model A	0.7681	0.6918	0.6945
Model B	0.6508	0.6611	0.6616

CONCLUSION

Model A demonstrated superior generalization on the final test set. Data augmentation in Model B resulted in a performance decrease, suggesting the base architecture is better suited for the raw input features in this specific setup.



Learning Curves



Training and validation curves for Model A and Model B

KEY TAKEAWAY

Model A achieved stronger final performance, while the curves also show some overfitting in the baseline model.

Final Evaluation Metrics

TEST ACCURACY

0.6945

MACRO PRECISION

0.7042

MACRO RECALL

0.6945

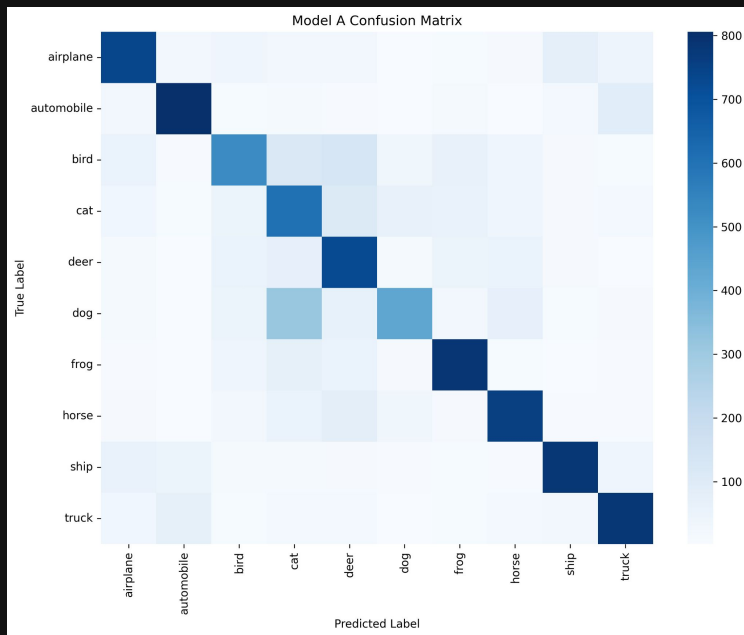
SELECTED MODEL

Model A

- Reasonably balanced multi-class performance
- Final model selected using evidence, not preference

Validation Method: k-fold Cross-Validation (k=5) • Metric Set: CIFAR-10
Test Partition

Confusion Matrix



INTERPRETATION

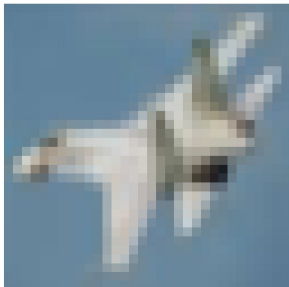
- Stronger performance on some vehicle classes such as ship and truck
- Several animal classes are harder to separate
- Low image resolution contributes to class confusion

Misclassified Examples

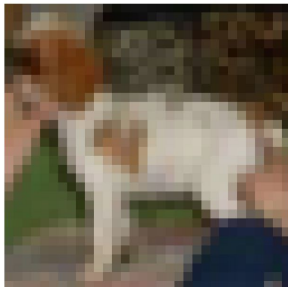
True: frog
Pred: deer



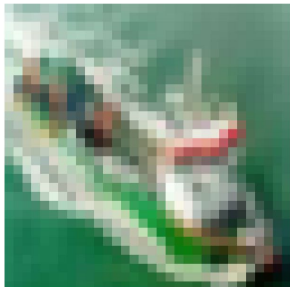
True: airplane
Pred: cat



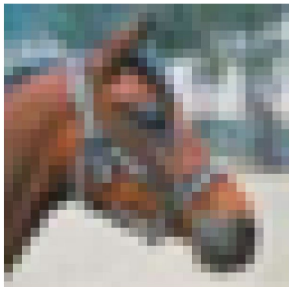
True: dog
Pred: cat



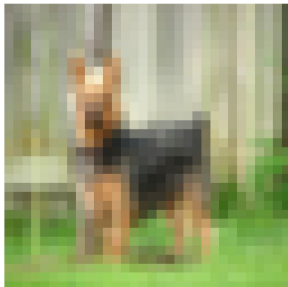
True: ship
Pred: frog



True: horse
Pred: deer



True: dog
Pred: deer



INTERPRETATION

- Several errors are understandable at 32 x 32 resolution
- Visually similar classes are harder to distinguish
- Misclassifications support the confusion matrix analysis

Comparative Positioning

COMPARED WITH CLASSICAL ML

- CNNs learn image features directly from raw pixels
- Less manual feature engineering is required
- Better suited to spatial patterns

COMPARED WITH ADVANCED DL

- Simpler than transfer learning or broader pipelines
- Easier to explain and justify academically
- Likely weaker in maximum performance

OVERALL JUDGEMENT

Appropriate and Defensible Approach

- Balances technical usefulness with clarity
- Valid solution for project scope
- Prioritizes explainability over complexity

Limitations and Lessons Learned

CRITICAL LIMITATIONS

- CIFAR-10 images are small and visually challenging
- Final model still shows some overfitting
- Limited number of model variations tested

CORE LESSONS LEARNED

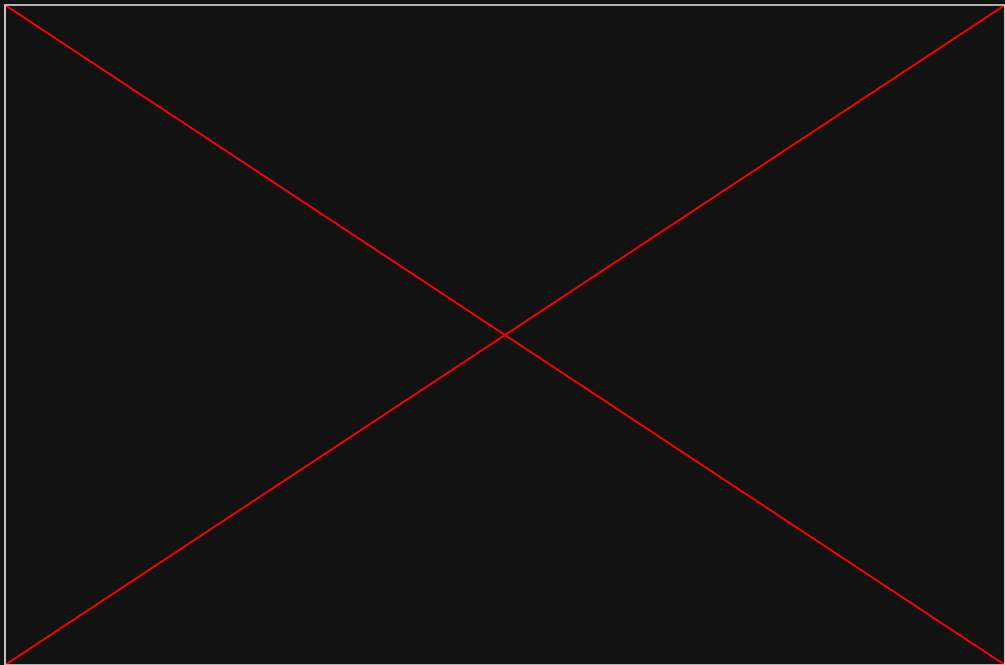
- Simple CNNs achieve meaningful recognition
- Architecture choices strongly affect results
- Data augmentation requires empirical testing

KEY RESEARCH TAKEAWAY

Model performance is a function of deliberate architectural iteration rather than sheer complexity.



Demo



Conclusion

Strategic Approach Summary

A CNN-based Track 2 approach was developed and evaluated on CIFAR-10. Model A was selected as the final candidate, as it consistently outperformed augmentation-based alternatives during empirical testing.

Test Accuracy

0.6945

Macro Precision

0.7042

Macro Recall

0.6945

Final Judgement

The approach is valid and defensible: careful evaluation matters as much as model construction.

References

Goodfellow, I., Bengio, Y. and Courville, A. (2016) *Deep Learning*. MIT Press.

Krizhevsky, A. (2009) *Learning multiple layers of features from tiny images*. Technical Report, University of Toronto.

LeCun, Y., Bottou, L., Bengio, Y. and Haffner, P. (1998) *Gradient-Based Learning Applied to Document Recognition*. Proceedings of the IEEE.

Shorten, C. and Khoshgoftaar, T.M. (2019) *A survey on Image Data Augmentation for Deep Learning*. Journal of Big Data.

TensorFlow (2024) `tf.keras.datasets.cifar10.load_data`. Google Open Source Documentation.

